

MU4MA329

Approximation of elliptic PDEs and numerical simulation Project

This project is an individual work. All implementation work should be done in Python 3.x and rely on the libraries written for scientific computing, namely NumPy, SciPy and Matplotlib. Some level of testing is expected in the implementation, and you will be specifically asked questions on the tests. All figures should be self-explanatory (label on every axis, legend, ...). This project is largely an adaptation of the code written during the programming sessions so that you should re-use some code already written.

1. **Convection-diffusion-reaction.** Let $\Omega := (0, 1)^2$, $f \in L^2(\Omega)$, $c > 0$ and $\mathbf{b} = (b_x, b_y) \in \mathbb{R}^2$. We consider the following second-order PDE model:

$$\begin{cases} \text{Find } u \in H^1(\Omega) \text{ such that :} \\ -\Delta u + \mathbf{b} \cdot \nabla u + cu = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases} \quad (1)$$

We want to compute an approximate solution u_h of the above problem using a conforming Galerkin method. The finite dimensional approximation space V_h is constructed using $\mathbb{P}_1$ Lagrange finite elements on triangular meshes.

- (a) Write the variational formulation associated to (1).
- (b) Write a routine `ElementaryConvectionMatrix` to assemble the elementary matrix associated to the convection term appearing in the bilinear form, on the model of what was done in TP3.
- (c) Write a routine `ConvectionMatrix` to assemble the full matrix for this term.
- (d) We consider the following source term

$$f(x, y) := e^{(b_x x + b_y y)/2} \sin(p\pi x) \sin(q\pi y), \quad \forall (x, y) \in \Omega, \quad (2)$$

for some $p, q \in \mathbb{N}$, $p, q > 0$.

Check that the analytic solution, denoted thereafter u_{ex} , to the problem (1) with the source term f given in (2) takes the form of $u_{\text{ex}} = \alpha f$, where you shall determine the value of $\alpha \in \mathbb{R}$.

- (e) Solve numerically (1) with the source term given in (2) for some $p, q \in \mathbb{N}$ and the values of the coefficients

$$\mathbf{b} = (1, 1), \quad c = 1. \quad (3)$$

- (f) Plot the numerical approximation u_h of the above problem.
- (g) Plot the convergence of the errors

$$\frac{\|u_h - \Pi_h u_{\text{ex}}\|_{L^2(\Omega)}}{\|\Pi_h u_{\text{ex}}\|_{L^2(\Omega)}} \quad \text{and} \quad \frac{\|u_h - \Pi_h u_{\text{ex}}\|_{H^1(\Omega)}}{\|\Pi_h u_{\text{ex}}\|_{H^1(\Omega)}}, \quad (4)$$

with respect to the mesh parameter h for various mesh refinements. What are the orders of the convergence?

2. **Connected components.** We consider now a domain Ω with $J > 0$ circular holes Ω_j , $j = 1, \dots, J$, see Figure 1. Let Γ_0 denote the exterior boundary, and $\Gamma_j := \partial\Omega_j$ the boundary of each hole, $j = 1, \dots, J$.

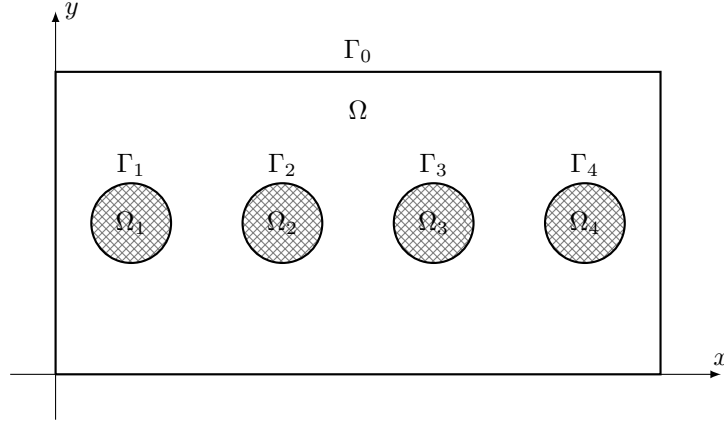


Figure 1: Domain geometry, $J = 4$.

- (a) Write a routine `ConnectedComponents` taking as input the connectivity array `bnd` $\in \mathbb{N}^{2 \times N_e^\partial}$ of the boundary of the mesh (output of the function `ExtractBoundary`) and giving as output an integer vector `labels` of size N_e^∂ such that `labels[k] = j` if the edge `bnd[:, k]` of the boundary belong to the boundary component Γ_j . Use the mesh `barwith4holes.msh`.
- (b) Adapt the routine `PlotMesh` to represent each connected component of the boundary of the mesh.
- (c) Solve the following problem on Ω

$$\begin{cases} \text{Find } u \in H^1(\Omega) \text{ such that :} \\ -\Delta u + \mathbf{b} \cdot \nabla u + cu = 0, & \text{in } \Omega, \\ u = j, & \text{on } \Gamma_j, j = 0, \dots, J. \end{cases} \quad (5)$$

Try increasing the modulus of the advection vector $\mathbf{b}$.